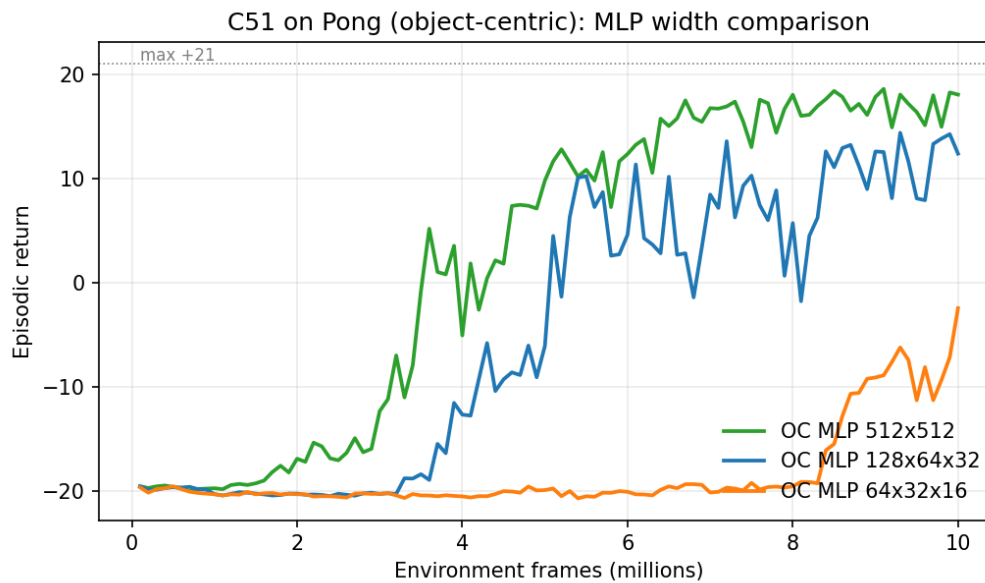


C51 on Pong (object-centric): MLP width comparison

Group 27 • Topic 27 • JAXAtari baselines

Three OC MLP widths, everything else held fixed (normalization, 10M frames, 8 envs, identical hyperparameters). Final episodic return at 10M frames:

OC network	Layers	Final return (10M)	Status at 10M
512×512	2	+18.0	solved, plateaued
$128 \rightarrow 64 \rightarrow 32$	3	+12.4	solved late, still rising
$64 \rightarrow 32 \rightarrow 16$	3	-2.4	mostly flat, steep unfinished climb



Decision

- Stick with **512×512** : fastest and highest within 10M, only choice that plateaus.
- $128 \rightarrow 64 \rightarrow 32$ is a reasonable middle ground but leaves return on the table at 10M.